

RC Upright Method

RC アップライト工法

White Paper v1.0

A Conceptual Study on the RC Upright Method

Proposal for an Adaptive and Disaster-Resilient Architectural Platform

更新型・災害適応型建築プラットフォームの提案

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RC Upright Method White Paper v1.0 — English

Proposal for an Adaptive and Disaster-Resilient Architectural Platform

1. Introduction

In recent years, Japanese society has experienced major changes in the environment surrounding architecture, including population decline, rising construction costs, vacant housing issues, intensified climate-related disasters, and shortages of construction labor.

Conventional architecture has generally been based on the concept of constructing and renewing buildings as a single integrated entity. Structural systems, equipment, interior finishes, and exterior appearance are often treated as having the same lifecycle. However, in reality, these elements have significantly different lifespans and social requirements. In rental housing especially, the renewability of equipment and exterior appearance strongly affects asset value and profitability.

This paper proposes the "RC Upright Method," which separates long-life reinforced concrete structural systems from renewable wooden components, thereby integrating partial renewability, post-disaster recoverability, flexible use conversion, and adaptability to the AI era.

2. Background of Development

The RC Upright Method was originally conceived approximately twenty years ago during the planning of a rental housing project. At that time, the client requested a rental housing system that could achieve low construction costs, high rental income, and long service life simultaneously.

Conventional lightweight steel-framed rental housing in nearby areas typically consisted of four units in a two-story configuration, with monthly rents of approximately ¥60,000 per unit.

In contrast, the proposed system adopted a townhouse-style configuration consisting of three units instead of four. Each unit was designed as an independent two-story residence with a loft, private garden space, and parking capacity for two vehicles.

This configuration achieved:

- Increased residential independence
- Detached-house-like living quality
- Higher market value
- Improved spatial flexibility

As a result, despite construction costs remaining roughly equivalent to conventional lightweight steel-framed apartments, the project achieved rents of approximately ¥110,000

per month per unit, generating significantly improved profitability.

3. Problems with Conventional Architecture

Conventional buildings are generally designed with structural systems distributed in both X and Y directions, minimizing eccentricity through balanced placement of columns, beams, and shear walls. As a result, buildings become structurally integrated systems, making large-scale renewal difficult without retaining major portions of the existing structure.

In rental housing particularly, profitability is strongly affected by:

- Exterior aging
- Equipment obsolescence
- Functional changes over time

However, conventional structural systems themselves often become obstacles to renewal. Additionally, conventional architecture tends to treat the entire building as having a single unified lifespan, despite the mismatch between structural durability and market lifecycle requirements.

4. Structural Philosophy of the RC Upright Method

The RC Upright Method utilizes long-life reinforced concrete wall structures as the primary structural support system, while renewable wooden structural components are inserted between them.

This concept applies the separation philosophy commonly found in civil engineering structures — such as the relationship between bridge piers and superstructures — to architectural systems. The objective is to separate long-life structural components from renewable and replaceable building elements.

From a structural mechanics perspective, adequate safety can be achieved as long as sufficient strength and deformation performance are secured in both X and Y directions. At the time of construction, advanced nonlinear dynamic seismic response simulations were conducted by an external professional engineering organization to verify the deformation performance of the reinforced concrete structural system.

5. Disaster Resilience

In recent years, climate change has intensified the risks of floods, tsunamis, typhoons, tornadoes, and other large-scale disasters. Conventional buildings often require complete demolition and reconstruction after severe flooding damage. The RC Upright Method improves disaster resilience through the use of reinforced concrete wall and floor systems.

5.1 Underfloor Flood Protection

For flood levels of approximately 60 cm or less, integrated flood protection systems can be installed together with the reinforced concrete wall structures.

5.2 Above-Floor Flood Recovery

Because the second-floor structure is supported by reinforced concrete structural elements, only the first-floor wooden components require removal and reconstruction after flooding. This allows continued use of the upper floor even during repair work.

5.3 Post-Disaster Reconstruction

Even in cases where flooding reaches the second floor, if the reinforced concrete walls and foundation remain intact, rapid recovery can be achieved through reconstruction of only the wooden structural components.

Flood Level	Response Capability
Up to ~60 cm (underfloor)	Integrated waterproofing system with RC walls
Above floor level (1st floor)	1st floor timber only — 2nd floor continues in use
Severe (2nd floor reached)	Timber reconstruction only if RC core intact

6. Characteristics as Renewable Architecture

As society evolves, building functions may need to change over time, including transitions between residential, commercial, office, warehouse, and garage uses.

In the RC Upright Method, each unit consists of an independent wooden structure that can be renewed individually from the façade side. Additionally, future openings can be pre-planned within reinforced concrete partition walls. These openings may initially be sealed with concrete blocks and later opened if larger integrated spaces become necessary.

Thus, the RC Upright Method views architecture not as a fixed object, but as a renewable and adaptable system.

7. Adaptability to the AI, AGI, and ASI Era

As artificial intelligence advances toward AGI (Artificial General Intelligence) and eventually ASI (Artificial Superintelligence), architecture is expected to increasingly incorporate:

- Automated construction

- Robotic assembly
- AI-assisted optimization
- Lifecycle management
- Modular construction systems

The RC Upright Method has strong compatibility with these future technologies due to its:

- Prefabrication potential
- Renewable structural configuration
- Modularity
- Partial replacement capability

Accordingly, the method can be regarded as an architectural platform highly adaptable to the AI era.

8. Future Vision

The RC Upright Method has the potential to contribute to:

- Diverse urban landscapes
- Renewable housing systems
- Climate-adaptive architecture
- Long-life building stock
- Advanced automated construction systems

Furthermore, future applications may include:

- Complex topographical sites
- Irregular land geometries
- High-rise structures
- Large-scale modular architecture

9. Conclusion

The RC Upright Method seeks to resolve the limitations of conventional architecture by separating long-life reinforced concrete structural systems from renewable wooden components.

Through this approach, the method improves renewability, disaster recoverability, flexibility of use, and lifecycle adaptability.

Furthermore, the method demonstrates strong compatibility with future AI-era construction systems, robotic assembly technologies, and lifecycle-oriented architectural management.

Therefore, the RC Upright Method should be regarded not merely as a construction method, but as a future-oriented renewable architectural platform.